STAFF REVIEW · H446 125 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. It passes a computed result to the caller.

2. B, since a stack uses LIFO.

3. Sum 0, carry-out 1.

4. It tracks the address of the next instruction to fetch.

I Do — source worked steps

1. Track each change — 00 INP: ACC = 7. — 01 STA 10: mailbox 10 becomes 7; ACC remains 7. — 02 LDA 10: ACC = 7. — 03 ADD 10: ACC = 14. — 04 OUT: output is 14 . — 05 HLT: stop. Data mailbox 10 is separate from instruction addresses 00–05.

We Do — guided CHECK

1. Predict the registers
What are ACC and mailbox 20 after the first two instructions?
Calculate ACC
Follow LDA 20 and SUB 21. Use the stored value 9 and constant 4.
Completed trace
INP sets ACC to 9. STA 20 saves 9. LDA 20 reloads it. SUB 21 subtracts 4, leaving ACC = 5. OUT prints 5 ; HLT stops.

You Do — source mark schemes

1. PC identifies next instruction to fetch [2]; accumulator holds data/arithmetic results [2]; mailboxes hold numbered instructions or data [2].

2. ACC=5 after INP [1]; mailbox 30 gets 5 [1]; LDA restores 5 [1]; ADD makes 8 [2]; OUT displays 8 [1].

3. One valid approach: INP; BRZ ZERO; OUT; HLT; ZERO OUT; HLT. BRZ branches if ACC=0 [2]; the nonzero path outputs entered number [2]; zero path outputs 0 [2]; HLT terminates and test values 0 and 7 exercise both paths [2].

Exit Ticket — source answer

1. False. STA stores ACC into memory. LDA reads a value from memory into ACC.